

Savitzky-Golay split command-lag search

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rotor median publish period [s]: 0.00499773025513
gimbal median publish period [s]: 0.00499868392944
initial lags [s]: {'rotor': 0.004997730255126953, 'gimbal': 0.004998683929443359}
smooth half-width multipliers: [4.0, 2.0, 1.0, 0.5]

smooth width=4: rotor=0.194308165s gimbal=0.0387508343s cost=1.50129604056
lag_gradient={'rotor': -4.943349679348286e-06, 'gimbal': -7.007268216230322e-07}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 23.44031072557584, 'finite_difference_l2_norm': 23.44031071220071,
'difference_l2_norm': 2.349868053175915e-07, 'relative_error': 1.0024901464347748e-08},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.022348123247527, 'finite_difference_l2_norm': 8.022348080844017,
'difference_l2_norm': 2.586169506105845e-07, 'relative_error': 3.2237064091142155e-08},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 17.189805820800284, 'finite_difference_l2_norm': 17.189805685842764,
'difference_l2_norm': 6.950240123410128e-07, 'relative_error': 4.04323364432662e-08}]
smooth width=2: rotor=0.192392485s gimbal=0.0386881261s cost=1.50268251572
lag_gradient={'rotor': 1.2151889297595808e-07, 'gimbal': 3.839846954012405e-06}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 26.34180426631433, 'finite_difference_l2_norm': 26.341803391809524,
'difference_l2_norm': 4.288675300180254e-06, 'relative_error': 1.6280871487852387e-07},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 8.741282842137668, 'finite_difference_l2_norm': 8.741282164764138,
'difference_l2_norm': 2.1909695163069442e-06, 'relative_error': 2.5064622159865333e-07},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 18.85920949625966, 'finite_difference_l2_norm': 18.859208190493728,
'difference_l2_norm': 4.503521829865532e-06, 'relative_error': 2.387969565086338e-07}]
smooth width=1: rotor=0.19508148s gimbal=0.0396850438s cost=1.50315949481 lag_gradient={'rotor':
-9.740668150537424e-05, 'gimbal': -8.189013752864238e-05} FD=[{'direction':
'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 16.182937640443694, 'finite_difference_l2_norm': 16.17487341886466,
'difference_l2_norm': 1.149018576062827, 'relative_error': 0.07100185402625848},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 10.55232606758503, 'finite_difference_l2_norm': 10.552320955830513,
'difference_l2_norm': 1.177998585399155e-05, 'relative_error': 1.1163402058033142e-06},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 21.225423799615854, 'finite_difference_l2_norm': 21.185699761216956,
'difference_l2_norm': 1.7866967447220203, 'relative_error': 0.08417720002153063}]
smooth width=0.5: rotor=0.195393364s gimbal=0.0393273044s cost=1.50313473049
lag_gradient={'rotor': 0.0009252338967714463, 'gimbal': -0.10464052221956924}
FD=[{'direction': 'negative_gradient', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 28.76081569581911, 'finite_difference_l2_norm': 28.835082550215404,
'difference_l2_norm': 4.161844981873935, 'relative_error': 0.14433268830169677},
{'direction': 'rotor_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 13.685777317061849, 'finite_difference_l2_norm': 13.685740032043583,
'difference_l2_norm': 7.999144981237753e-05, 'relative_error': 5.844859810239161e-06},
{'direction': 'gimbal_delay_seconds', 'scheme': 'central', 'step': 6.0554544523933395e-06,
'analytic_l2_norm': 28.755064877785763, 'finite_difference_l2_norm': 28.82921469540368,
'difference_l2_norm': 4.162074405697035, 'relative_error': 0.1443700235914024}]

strict profile iterations: 1
iteration 1: best=(0.195393363886, 0.039327304411) cost=1.50389032917
rotor_range=(0.190395633631, 0.2) gimbal_range=(0.034328620481, 0.04432598834) expansions=[]
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selected rotor lag 0.195393363886 s (39.0964 publish periods)
selected gimbal lag 0.039327304411 s (7.86753 publish periods)
selected objective cost 1.50389008473